Rational rotations of three dimensional Euclidean space

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Abstract

We give an algebraic and projective account of rotations of three dimensional space over a field of characteristic not two. No angles, circular functions or unit-vector normalizations are required. On the affine chart the main objects are the **half spread**, which measures the amount of a rotation, and the **anchor**, an antisymmetric matrix which determines every rotation except a half turn. The associated **cap** is a Cayley transform. Passing from anchors to quaternion proportions removes the exceptional cases: the non-null points of $P^3(\mathbb{F})$ form a group which maps bijectively and multiplicatively onto $SO(3,\mathbb{F})$. Over $\mathbb{Q}$ every projective point is non-null, so $SO(3,\mathbb{Q})$ is $P^3(\mathbb{Q})$ equipped with the projective quaternion product. The square class of the quaternion quadrance is the spinor norm. It is onto $\mathbb{Q}^+/(\mathbb{Q}^*)^2$, and consequently rational unit quaternions reach only a subgroup of infinite index of $SO(3,\mathbb{Q})$. Quaternion rays nevertheless give a two–to–one cover of all rational rotations; over the real numbers this reduces to the familiar $SU(2) = Spin(3)$ picture. A 2×2 matrix model makes the connection with Pauli matrices explicit and places the Euclidean theory beside the corresponding algebraic theory of rotations and boosts in three dimensional relativistic geometry, developed in a companion paper.

1 Introduction

Rotations occur throughout geometry and physics. They describe rigid motion, attitude, angular momentum, molecular and crystal orientation, robotics and computer graphics. Their standard descriptions use rotation matrices, Euler angles, Rodrigues parameters or unit quaternions. These are closely related, but the usual presentation mixes algebraic operations with angles, trigonometric functions and normalizations.

There is a simpler algebraic core. It goes back to Euler and especially to Rodrigues [1], [5]. Rodrigues found both an explicit formula for a rotation and a composition law for the corresponding parameters, before Hamilton introduced quaternions [12], [10]. The Cayley transform and quaternion algebra later put these formulas into a broader setting [6], [8], [11].

This paper develops that algebraic core over a general field $\mathbb{F}$ of characteristic not two. The rational field $\mathbb{Q}$ is our main example. We use addition, multiplication and division, together with projective proportions. Angles and circular functions are not part of the definitions. In this respect the paper continues the program of rational trigonometry [13], [14], in which quadrance and spread replace distance and angle.

Three objects organize the affine theory. The **half spread** p measures the amount of a rotation. The **anchor** is an antisymmetric matrix, or equivalently a vector, attached to every rotation which is not a half turn. The **cap** of an anchor A,

$$c(A) = (I + A)(I - A)^{-1},$$

recovers the rotation from the anchor. Composition of rotations then becomes a rational operation on anchor vectors.

The affine description is useful but incomplete. Half turns have no finite anchor. The anchor product has a forbidden denominator. Its associativity is therefore only conditional. Over fields with null vectors there is also an excluded quadric of anchors. All of these effects come from using one affine chart of a projective object.

The projective object is a quaternion proportion

$$(x : v) = (x : t : r : s).$$

For a non-null proportion we define

$$\rho(x : v) = I + \frac{2(xA_v + A_v^2)}{x^2 + Q(v)}.$$

Here $Q(v) \equiv t^2 + r^2 + s^2$, and A_v is the anchor of $v = (t, r, s)$, the antisymmetric matrix defined in Section 3. The main theorem says that these proportions give all rotations, uniquely, and that quaternion multiplication gives composition. Over $\mathbb{Q}$ the quadratic form

$$x^2 + t^2 + r^2 + s^2$$

has no non-zero null vectors. Thus every point of $P^3(\mathbb{Q})$ represents a rational rotation.

Here is a first example. Take

$$(x : t : r : s) = (2 : 1 : 2 : 3).$$

Then

$$\rho(2 : 1 : 2 : 3) = \frac{1}{9}\begin{pmatrix} -4 & 8 & -1 \\ -4 & -1 & 8 \\ 7 & 4 & 4 \end{pmatrix}.$$

The rows are mutually perpendicular, each has quadrance 1, and the determinant is 1. The vector $(1, 2, 3)$ is fixed, so it gives the axis. The half spread is

$$p = \frac{1^2 + 2^2 + 3^2}{2^2 + 1^2 + 2^2 + 3^2} = \frac{7}{9},$$

and this is also

$$p = \frac{3 - \operatorname{tr}\rho}{4}.$$

The inverse is represented by $(2 : -1 : -2 : -3)$. Composition with another rotation requires only the four component quaternion product. No normalization is involved.

The classical dictionary

For comparison only, suppose that $\mathbb{F} = \mathbb{R}$ and that the rotation is described classically by a unit axis n and an angle θ. With our sign conventions one may choose the quaternion proportion

$$\left(\cos\frac{\theta}{2} : n\sin\frac{\theta}{2}\right).$$

On the affine chart where $\cos\frac{\theta}{2} \neq 0$, the anchor vector is

$$v = n\tan\frac{\theta}{2}.$$

The half spread and the rotation spread are

$$p = \sin^2\frac{\theta}{2}, \qquad s = 4p\,(1-p) = \sin^2\theta,$$

and the anchor quadrance is

$$Q\,(v) = \tan^2\frac{\theta}{2} = \frac{p}{1-p}.$$

These identities connect the present theory with the usual one. They are not definitions here. The algebraic quantities on the right continue to make sense over $\mathbb{Q}$, finite fields and other computational fields where the angle θ has no role.

This dictionary is also useful in physics. Over $\mathbb{R}$ a non-zero quaternion can be normalized to unit quadrance, and the unit quaternions give $SU\,(2) = Spin\,(3)$, with the familiar two–to–one map onto $SO\,(3,\mathbb{R})$ [11]. Over $\mathbb{Q}$ this normalization usually leaves the rational field. The projective theory does not. In fact the rational unit quaternions reach only the kernel of a square class homomorphism, the spinor norm, and that kernel has infinite index in $SO\,(3,\mathbb{Q})$. Quaternion rays still give a two–to–one cover of all rational rotations. This distinction between normalization and projectivization is one of the main arithmetic points of the paper.

There is a parallel relativistic theory. Replacing the positive quadratic form by one of signature $(1,2)$ leads to the rotations and boosts of three dimensional Minkowski geometry. The corresponding four dimensional algebra is a 2×2 matrix algebra rather than the Euclidean quaternion division algebra over $\mathbb{Q}$. The same projective ideas, including null directions and parabolic transformations, continue to apply. We develop that theory in the companion paper [17]. The matrix model of Section 11.3 displays the Euclidean and relativistic theories side by side.

The paper proceeds as follows. Sections 2 and 3 develop the vector and anchor algebra. Section 4 treats reflections and half turns. Sections 5–7 introduce the cap, half spread and anchor of a rotation. Section 8 gives an algebraic orientation test, and Section 9 gives the anchor composition law. Section 10 isolates the limits of the affine chart. Section 11 passes to quaternion proportions and proves the projective rotation theorem. Section 12 studies the spinor-norm obstruction over $\mathbb{Q}$.

2 Quadrances, spreads and vector algebra

Throughout, $\mathbb{F}$ is a field of characteristic not two. The rational case $\mathbb{F} = \mathbb{Q}$ is primary, and finite fields form the other main family of examples. Vectors are row vectors

$$v \equiv (t, r, s) \in \mathbb{F}^3$$

with the **dot product**

$$v_1 \cdot v_2 \equiv v_1 v_2^T = t_1 t_2 + r_1 r_2 + s_1 s_2$$

and the **cross product**

$$v_1 \times v_2 \equiv (r_1 s_2 - r_2 s_1, s_1 t_2 - s_2 t_1, t_1 r_2 - t_2 r_1).$$

Vectors v_1 and v_2 are **perpendicular** exactly when $v_1 \cdot v_2 = 0$. For a non-zero vector v we write $[v]$ for the **line** it spans, the set of scalar multiples of v.

Definition 1 *The **quadrance** of the vector $v = (t, r, s)$ is*

$$Q(v) \equiv v \cdot v = t^2 + r^2 + s^2.$$

*The vector v is **null** exactly when $Q(v) = 0$.*

The zero vector is always null. Other null vectors may or may not exist. This distinction will matter throughout the paper.

Theorem 2 (Null vectors) *Over an ordered field, and in particular over $\mathbb{Q}$, the only null vector is 0. Over any finite field of odd characteristic there exist non-zero null vectors.*

Proof. Over an ordered field a sum of squares vanishes only when each term does. Over a finite field $\mathbb{F}_q$ with q odd, the squares number $(q+1)/2$ including 0, so the sets $\{t^2 : t \in \mathbb{F}_q\}$ and $\{-1 - r^2 : r \in \mathbb{F}_q\}$, each of size $(q+1)/2$, must intersect; any common value gives $t^2 + r^2 + 1 = 0$ and hence the non-zero null vector $(t, r, 1)$. ■

Example 3 *Over $\mathbb{F}_5$ the vector $(1, 2, 0)$ is null, since $1 + 4 = 0$. Note that a non-zero null vector is perpendicular to itself.*

Definition 4 *The **spread** between non-null vectors v_1 and v_2 is*

$$s(v_1, v_2) \equiv 1 - \frac{(v_1 \cdot v_2)^2}{Q(v_1) Q(v_2)}.$$

The spread is unchanged when v_1 or v_2 is multiplied by a non-zero scalar, so it is a quantity attached to the pair of lines $[v_1]$ and $[v_2]$. For non-null vectors, $s(v_1, v_2) = 1$ exactly when v_1 and v_2 are perpendicular.

The vector identities needed below are polynomial identities in the coordinates, so they hold over any field. We record them here.

Theorem 5 (Vector algebra) *For any vectors u, v, w, z in $\mathbb{F}^3$:*

1. *(Scalar triple product)* $(u \times v) \cdot w = \det \begin{pmatrix} u \\ v \\ w \end{pmatrix}$, *and in particular* $(u \times v) \cdot u = (u \times v) \cdot v = 0$;
2. *(Vector triple product)* $u \times (v \times w) = v\,(u \cdot w) - w\,(u \cdot v)$;
3. *(Binet–Cauchy identity)* $(u \times v) \cdot (w \times z) = (u \cdot w)\,(v \cdot z) - (u \cdot z)\,(v \cdot w)$.

The same algebra extends to any non-degenerate symmetric bilinear form on $\mathbb{F}^3$, with the corresponding generalised vector product; see [15], [16]. Here we use only the Euclidean form.

Theorem 6 (Lagrange identity) *For any vectors v_1 and v_2,*

$$(v_1 \cdot v_2)^2 + Q\,(v_1 \times v_2) = Q\,(v_1)\,Q\,(v_2)$$

and consequently for non-null v_1, v_2

$$s\,(v_1, v_2) = \frac{Q\,(v_1 \times v_2)}{Q\,(v_1)\,Q\,(v_2)}.$$

Proof. The Lagrange identity is the case $w = v_1, z = v_2$ of the Binet–Cauchy identity. The spread formula follows by dividing by $Q\,(v_1)\,Q\,(v_2)$. ∎

Two consequences will be useful. If v_1 and v_2 are non-null but $v_1 \times v_2$ is non-zero and null, then $s\,(v_1, v_2) = 0$ although the two lines are distinct. Thus zero spread need not mean equality of lines over a field with null vectors. Over an ordered field the spread of two non-zero vectors lies between 0 and 1.

We also need the following elementary polynomial principle.

Lemma 7 (Vanishing lemma) *Let g and h be polynomials in finitely many variables with rational coefficients, with h not the zero polynomial. If g vanishes at every rational point at which h does not vanish, then g is the zero polynomial.*

Proof. The product gh vanishes at every rational point. A polynomial over an infinite field which vanishes at every point is the zero polynomial, by induction on the number of variables, since a non-zero polynomial in one variable has only finitely many roots. So $gh = 0$, and since the polynomial ring has no zero divisors and $h \neq 0$, $g = 0$. ∎

Hence an integral polynomial identity proved on a non-empty rational generic locus is an identity over every field.

3 Anchors

We now encode vectors by antisymmetric matrices.

Definition 8 *The **anchor** of the vector $v = (t, r, s)$ is the matrix*

$$A_v \equiv \begin{pmatrix} 0 & s & -r \\ -s & 0 & t \\ r & -t & 0 \end{pmatrix}.$$

The map $v \mapsto A_v$ is a linear bijection from $\mathbb{F}^3$ onto the space

$$so\,(3) \equiv \left\{A : A^T = -A\right\}$$

of antisymmetric 3×3 matrices. The anchor acts on vectors as a cross product:

Lemma 9 (Cross product lemma) *For any vectors v and w,*

$$wA_v = v \times w.$$

In particular $vA_v = 0$.

Proof. Both sides are bilinear in (v, w), so it suffices to check the nine cases where v and w are standard basis vectors, which is immediate. ∎

For row vectors v and w we write $w^T v$ for the 3×3 matrix with entries $\left(w^T v\right)_{ij} = w_i v_j$. The main anchor identity is the following.

Theorem 10 (Anchor product identity) *For any vectors v and w,*

$$A_v A_w = w^T v - (v \cdot w)\, I.$$

Proof. For any vector x, the cross product lemma and the vector triple product identity of Section 2 give

$$x\,(A_v A_w) = (v \times x)\, A_w = w \times (v \times x) = v\,(w \cdot x) - x\,(w \cdot v) = x\left(w^T v - (v \cdot w)\, I\right)$$

and two matrices agreeing on all row vectors x are equal. ∎

Note the reversal of order between the left and right sides. The identity has the following immediate consequences.

Corollary 11 *For any vectors v and w:*

1. $A_v^2 = v^T v - Q\,(v)\, I$;
2. $A_v A_w - A_w A_v = w^T v - v^T w = A_{w \times v}$;
3. $A_v A_w A_v = -\,(v \cdot w)\, A_v$;
4. $\operatorname{tr}\,(A_v A_w) = -2\,(v \cdot w)$.

Proof. For 1, set $w = v$ in the identity. For 2, subtract the identity from its transpose–swap; that the antisymmetric matrix $w^T v - v^T w$ is the anchor of $w \times v$ is checked on basis vectors, or by applying the cross product lemma and the vector triple product identity once more. For 3,

$$A_v A_w A_v = \left(w^T v - (v \cdot w)\, I\right) A_v = w^T\,(vA_v) - (v \cdot w)\, A_v = -\,(v \cdot w)\, A_v$$

using $vA_v = 0$. For 4, $\operatorname{tr}\left(w^T v\right) = v \cdot w$, so the trace of the identity is $(v \cdot w) - 3\,(v \cdot w)$. ∎

Part 4 gives a natural bilinear form on the anchor space.

Definition 12 *The **anchor form** of anchors A and B is*

$$A \cdot B \equiv -\frac{1}{2} \operatorname{tr}\,(AB)\,,$$

*and the **quadrance** of an anchor is $Q\,(A) \equiv A \cdot A$.*

Theorem 13 (Isometry) *For any vectors v and w,*

$$A_v \cdot A_w = v \cdot w \qquad \text{and} \qquad Q(A_v) = Q(v),$$

so $v \mapsto A_v$ is an isometry between $\mathbb{F}^3$ with the dot product and $so(3)$ with the anchor form.

Theorem 14 (Characteristic polynomial of an anchor) *The characteristic polynomial of A_v is*

$$\det(XI - A_v) = X\left(X^2 + Q(v)\right) = X^3 + Q(v)X.$$

Proof. The coefficient of X^2 is $-\operatorname{tr} A_v = 0$; the coefficient of X is the sum of the principal 2×2 minors, namely $s^2 + r^2 + t^2 = Q(v)$; and $\det A_v = 0$ since A_v is an antisymmetric matrix of odd size:

$$\det A_v = \det A_v^T = \det(-A_v) = (-1)^3 \det A_v = -\det A_v$$

so that $2 \det A_v = 0$, and the characteristic is not two. ■

4 Reflections and half turns

A **rotation** is a 3×3 matrix ρ over $\mathbb{F}$ with $\rho\rho^T = I$ and $\det \rho = 1$. Since $(w_1\rho)\cdot(w_2\rho) = w_1\rho\rho^T w_2^T = w_1 \cdot w_2$, rotations preserve the dot product. The rotations form a group $SO(3, \mathbb{F})$, also written $SO(3)$; the matrices satisfying only $\rho\rho^T = I$ form the group $O(3)$.

Definition 15 *For a non-null vector v, the **reflection** in v is the matrix*

$$\sigma_v \equiv I - \frac{2}{Q(v)} v^T v.$$

Since $\sigma_{\lambda v} = \sigma_v$ for any non-zero scalar λ, the reflection depends only on the line $[v]$. Geometrically σ_v is the reflection in the plane perpendicular to v: this is the content of parts 2 and 3 of the following.

Theorem 16 (Reflections) *For a non-null vector v:*

1. *σ_v is symmetric and $\sigma_v^2 = I$;*
2. *$v\sigma_v = -v$;*
3. *$w\sigma_v = w$ exactly when $w \cdot v = 0$;*
4. *$\sigma_v \in O(3)$;*
5. *$\det \sigma_v = -1$.*

Proof. Symmetry is visible. Since $\left(v^T v\right)\left(v^T v\right) = v^T\left(v v^T\right) v = Q(v)\, v^T v$,

$$\sigma_v^2 = I - \frac{4}{Q(v)} v^T v + \frac{4}{Q(v)^2} Q(v)\, v^T v = I$$

which gives 1, and 4 follows since $\sigma_v^T = \sigma_v$. For 2 and 3, $w\sigma_v = w - \frac{2(w\cdot v)}{Q(v)} v$, which is $-v$ when $w = v$ and is w exactly when $w \cdot v = 0$. For 5: since v is non-null, v does not lie in the plane

$v^{\perp} \equiv \{w : w \cdot v = 0\}$, which has dimension two; choosing a basis of $\mathbb{F}^3$ consisting of v and a basis of $v^{\perp}$, the matrix of σ_v is diagonal with entries $-1, 1, 1$ by 2 and 3. ■

Note the essential role of non-nullity: for a non-zero null vector v the expression defining σ_v is meaningless, and indeed v lies in its own perpendicular plane, so there is no reflection interchanging the sides of that plane along v.

Definition 17 *For a non-null vector v, the **half turn** about the line $[v]$ is*

$$\rho_{[v]} \equiv -\sigma_v = -I + \frac{2}{Q(v)} v^T v.$$

Theorem 18 (Half turns) *For a non-null vector v: the half turn $\rho_{[v]}$ lies in $SO(3)$, fixes v, negates every vector perpendicular to v, and satisfies $\rho_{[v]}^2 = I$. It is the unique 3×3 matrix which fixes v and negates every vector perpendicular to v, and its fixed vectors are exactly the vectors of $[v]$ together with 0. Consequently distinct non-null lines have distinct half turns.*

Proof. The first assertions follow from the previous theorem after multiplication by $-I$, using $\det(-\sigma_v) = (-1)^3 \det \sigma_v = 1$. As in the proof of part 5 there, $\mathbb{F}^3$ is the direct sum of $[v]$ with $v^{\perp}$, so a matrix is determined by its action on v and on $v^{\perp}$, which gives uniqueness. Writing any vector as $w = \lambda v + w'$ with $w' \in v^{\perp}$, we have $w\rho_{[v]} = \lambda v - w'$, which equals w precisely when $2w' = 0$, that is $w' = 0$. So the line $[v]$ is recovered from $\rho_{[v]}$. ■

Two reflections are enough to produce every rotation.

Theorem 19 (Two reflections theorem) *Every rotation $\rho \in SO(3, \mathbb{F})$ is a product*

$$\rho = \sigma_{v_1} \sigma_{v_2}$$

of two reflections in non-null vectors, and conversely every such product is a rotation. The factorization is not unique.

The converse direction is immediate: a product of two elements of $O(3)$ each of determinant -1 lies in $SO(3)$. The direct statement is proved in Section 7.2, where the factorization is constructed from the anchor of ρ and its freedom is made precise.

Theorem 20 (Axis of a composition) *If v_1 and v_2 are non-null and independent, then the rotation $\sigma_{v_1}\sigma_{v_2}$ fixes every vector on the line $[v_1 \times v_2]$.*

Proof. The vector $v_1 \times v_2$ is perpendicular to both v_1 and v_2 by the scalar triple product identity, so it is fixed by both reflections. ■

Example 21 *Let $v_1 = (1, 2, 3)$ and $v_2 = (2, 1, -1)$, with quadrances 14 and 6. Then*

$$\sigma_{v_1} = \frac{1}{7} \begin{pmatrix} 6 & -2 & -3 \\ -2 & 3 & -6 \\ -3 & -6 & -2 \end{pmatrix}, \qquad \sigma_{v_2} = \frac{1}{3} \begin{pmatrix} -1 & -2 & 2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}$$

and

$$\sigma_{v_1}\sigma_{v_2} = \frac{1}{21} \begin{pmatrix} -8 & -19 & 4 \\ -16 & 4 & -13 \\ 11 & -8 & -16 \end{pmatrix} \in SO(3)$$

with fixed vector $v_1 \times v_2 = (-5, 7, -3)$: indeed $(-5, 7, -3)\,\sigma_{v_1}\sigma_{v_2} = (-5, 7, -3)$.

5 The cap of an anchor

Definition 22 *The **cap** of an anchor A is the matrix*

$$c(A) \equiv (I + A)(I - A)^{-1}$$

whenever $I - A$ is invertible.

This is a form of the classical Cayley transform [6]. From the characteristic polynomial of A_v we get

$$\det(I - A_v) = \det(I + A_v) = 1 + Q(v)$$

Thus the cap of A_v is defined exactly when $Q(v) \neq -1$. Over $\mathbb{Q}$, and over any ordered field, every anchor has a cap. Over a finite field the quadric $Q = -1$ is excluded. This will fit naturally into the projective picture.

The inverse of $I - A$ reduces to a quadratic polynomial in A:

Lemma 23 *If $Q \equiv Q(v) \neq -1$ and $A \equiv A_v$ then*

$$(I - A)^{-1} = I + \frac{A + A^2}{1 + Q}.$$

Proof. Using $A^3 = -QA$ from the characteristic polynomial,

$$(I - A)\left((1 + Q)I + A + A^2\right) = (1 + Q)I + A + A^2 - (1 + Q)A - A^2 - A^3 = (1 + Q)I.$$

■

Theorem 24 (Quadratic reduction) *If $Q \equiv Q(v) \neq -1$ and $A \equiv A_v$ then*

$$c(A) = I + \frac{2}{1 + Q}\left(A + A^2\right).$$

Proof. By the lemma,

$$c(A) = (I + A)\left(I + \frac{A + A^2}{1 + Q}\right) = I + A + \frac{A + 2A^2 + A^3}{1 + Q} = I + A + \frac{(1 - Q)A + 2A^2}{1 + Q} = I + \frac{2(A + A^2)}{1 + Q}.$$

■

In coordinates, with $v = (t, r, s)$ and $Q = t^2 + r^2 + s^2$:

$$c(A_v) = \frac{1}{1 + Q}\begin{pmatrix} t^2 - r^2 - s^2 + 1 & 2(s + rt) & 2(-r + st) \\ 2(-s + rt) & -t^2 + r^2 - s^2 + 1 & 2(t + rs) \\ 2(r + st) & 2(-t + rs) & -t^2 - r^2 + s^2 + 1 \end{pmatrix}.$$

Theorem 25 (Caps are rotations) *If $Q(v) \neq -1$ and $\rho \equiv c(A_v)$ then:*

1. *$\rho \in SO(3)$;*
2. *$v\rho = v$;*
3. *the characteristic polynomial of ρ is*

$$(X-1)\left(X^2 - 2\frac{1-Q}{1+Q}X + 1\right);$$

4. *-1 is not an eigenvalue of ρ, so no cap is a half turn;*
5. *the cap is injective: $c(A)$ determines A.*

Proof. For 1: the four matrices $I+A$, $(I-A)^{-1}$, $(I+A)^{-1}$, $I-A$ are polynomials in A by the lemma, hence commute, and since $(I+A)^T = I - A$,

$$\rho\rho^T = (I+A)(I-A)^{-1}(I+A)^{-1}(I-A) = I,$$

while $\det\rho = \det(I+A)/\det(I-A) = 1$. For 2: $vA = 0$ gives $v(I+A) = v$ and $v(I-A)^{-1} = v$. For 3: by 1 the characteristic polynomial has the form $(X-1)(X^2 - \alpha X + 1)$ with $\alpha = \operatorname{tr}\rho - 1$, and by quadratic reduction

$$\operatorname{tr}\rho = 3 + \frac{2}{1+Q}\left(\operatorname{tr}A + \operatorname{tr}A^2\right) = 3 + \frac{2(0-2Q)}{1+Q} = \frac{3-Q}{1+Q}$$

using $\operatorname{tr}A^2 = -2Q$ from the Isometry theorem. For 4: evaluating the quadratic factor at $X = -1$ gives $2 + 2\frac{1-Q}{1+Q} = \frac{4}{1+Q} \neq 0$. For 5: the map $w \mapsto (I+w)(I-w)^{-1}$ is inverted by $z \mapsto (z-I)(z+I)^{-1}$, which is defined on ρ by 4. ∎

Example 26 *Let $v = (4,1,7)$, so $Q = 66$ and $1+Q = 67$. Then*

$$\rho = c(A_v) = \frac{1}{67}\begin{pmatrix} -33 & 22 & 54 \\ -6 & -63 & 22 \\ 58 & 6 & 33 \end{pmatrix}$$

with $(4,1,7)\rho = (4,1,7)$ and characteristic polynomial $(X-1)\left(X^2 + \frac{130}{67}X + 1\right)$.

Example 27 *Over $\mathbb{F}_5$ the vector $v = (2,0,0)$ has $Q(v) = 4 = -1$, so $c(A_v)$ is not defined: the matrix $I - A_v$ has determinant 0. The quadric of such anchors is invisible over $\mathbb{Q}$ but unavoidable over finite fields.*

Example 28 *A null anchor* does *have a cap, and it is unipotent: if $Q(v) = 0$ then $c(A_v) = I + 2(A_v + A_v^2)$ with characteristic polynomial $(X-1)^3$. Over $\mathbb{F}_5$ with the null vector $v = (1,2,0)$ of Section 2,*

$$c(A_v) = \begin{pmatrix} 3 & 4 & 1 \\ 4 & 4 & 2 \\ 4 & 3 & 1 \end{pmatrix}$$

is a rotation fixing v, all of whose eigenvalues are 1, yet which is not the identity. Such parabolic rotations are the analogues, inside the Euclidean theory over $\mathbb{F}_5$, of the null rotations of relativistic geometry; over $\mathbb{Q}$ they do not exist.

The cap misses every half turn. Over some fields it also misses the quadric $Q = -1$ in its domain. These are features of the affine chart. Section 11 removes them by projectivization. We first finish the affine theory.

6 Half spreads

Definition 29 *The **half spread** of a rotation $\rho \in SO(3)$ is*

$$p(\rho) \equiv \frac{3 - \operatorname{tr}\rho}{4}.$$

Lemma 30 (Fixed direction) *Every rotation $\rho \in SO(3, \mathbb{F})$ has 1 as an eigenvalue.*

Proof. Transposing, then using $\rho^T = \rho^{-1}$, $\rho^{-1} - I = \rho^{-1}(I - \rho)$ and $\det \rho^{-1} = 1$,

$$\det(\rho - I) = \det\left(\rho^T - I\right) = \det\left(\rho^{-1} - I\right) = \det \rho^{-1} \det(I - \rho) = (-1)^3 \det(\rho - I).$$

The characteristic is not two, so $\det(\rho - I) = 0$. ■

Theorem 31 (Half spread identities) *If $\rho = c(A_v)$ with $Q \equiv Q(v) \neq -1$, then $p \equiv p(\rho)$ satisfies*

$$p = \frac{Q}{1+Q}, \qquad 1 - p = \frac{1}{1+Q}, \qquad 1 - 2p = \frac{1-Q}{1+Q}, \qquad p(1-p) = \frac{Q}{(1+Q)^2}$$

and $p \neq 1$, with the quadrance recovered as

$$Q = \frac{p}{1-p}.$$

The characteristic polynomial of any rotation ρ is

$$(X - 1)\left(X^2 - 2(1 - 2p(\rho))X + 1\right).$$

If ρ is a half turn then $p(\rho) = 1$.

Proof. From Section 5, $\operatorname{tr} c(A_v) = \frac{3-Q}{1+Q}$. Hence $p = \frac{Q}{1+Q}$, and the other displayed identities follow immediately. The previous lemma gives the factor $X - 1$ in the characteristic polynomial. Since $\det \rho = 1$, the remaining quadratic has constant term 1, so it is $X^2 - \alpha X + 1$. Its coefficient is $\alpha = \operatorname{tr}\rho - 1 = 2 - 4p = 2(1 - 2p)$. A half turn has trace -1, and therefore $p = 1$. ■

Thus the half spread separates caps from half turns: every rotation with $p \neq 1$ has an anchor (Section 7), and the quadrance of that anchor is the **half twist** $p/(1-p)$. Over an ordered field $p = \frac{Q}{1+Q}$ lies in the interval $[0, 1]$; over a finite field every value of p occurs, since every element of a finite field is a sum of three squares.

Definition 32 *The **spread** of the rotation $\rho = c(A_v)$, or $\rho = \rho_{[v]}$, is*

$$s(\rho) \equiv s(w, w\rho)$$

for any non-null vector w perpendicular to v.

Theorem 33 (Spread of a rotation) *The spread is well defined, independent of the choice of w, and*

$$s(\rho) = 4p(1-p) \qquad \textit{where} \qquad p = p(\rho).$$

Proof. Let $\rho = c(A_v)$, $Q \equiv Q(v)$, and let w be non-null with $w \cdot v = 0$. Then $wA_v = v \times w$ and $wA_v^2 = (w \cdot v)\, v - Qw = -Qw$, so by quadratic reduction

$$w\rho = \frac{(1-Q)\, w + 2\,(v \times w)}{1+Q}.$$

Since $(v \times w) \cdot w = 0$ and $Q(w\rho) = Q(w)$,

$$s(w, w\rho) = 1 - \frac{(w \cdot w\rho)^2}{Q(w)\, Q(w\rho)} = 1 - \frac{(1-Q)^2}{(1+Q)^2} = \frac{4Q}{(1+Q)^2} = 4p(1-p).$$

For a half turn, $w\rho = -w$ and $s(w, -w) = 0 = 4 \cdot 1 \cdot (1-1)$. Finally such a w exists: the restriction of the dot product to the plane perpendicular to a non-null v is non-degenerate, and a non-degenerate binary form represents non-zero values. ∎

The polynomial $S_2(p) \equiv 4p(1-p)$ is the second spread polynomial of rational trigonometry [13]. It also governs the squaring of a rotation, which is the rational form of the doubling of angles.

Theorem 34 (Half spread of a square) *For any rotation ρ with half spread p,*

$$p\left(\rho^2\right) = 4p(1-p).$$

In particular the half spread of ρ^2 is the spread of ρ.

Proof. By the half spread identities the characteristic polynomial of ρ is $(X-1)\left(X^2 - \alpha X + 1\right) = X^3 - (1+\alpha) X^2 + (1+\alpha) X - 1$, so the trace e_1 of ρ and the sum e_2 of its principal 2×2 minors are both equal to $1 + \alpha = \operatorname{tr}\rho = 3 - 4p$. By Newton's identity $\operatorname{tr}\left(\rho^2\right) = e_1^2 - 2e_2$, valid for any 3×3 matrix,

$$\operatorname{tr}\left(\rho^2\right) = (3-4p)^2 - 2(3-4p) = 3 - 16p + 16p^2$$

and so $p\left(\rho^2\right) = \frac{3 - \operatorname{tr}\left(\rho^2\right)}{4} = 4p - 4p^2$. ∎

Note that the spread does not determine the amount of rotation: the identity and every half turn all have spread 0. The half spread does better, distinguishing the identity ($p = 0$) from the half turns ($p = 1$); though over a field with non-zero null vectors even the half spread has degenerate directions, as the parabolic rotations of Section 5 show: they satisfy $p = 0$ without being the identity. Over $\mathbb{Q}$ no such degeneracy occurs, and we will see in Sections 8 and 12.1 that a rotation is completely determined by its axis, its half spread, and one further bit of information, its handedness.

Theorem 35 (Spread of normals) *For non-null vectors v_1 and v_2,*

$$p\left(\sigma_{v_1}\sigma_{v_2}\right) = s(v_1, v_2).$$

Proof. Since $\operatorname{tr}\left(v^T v\right) = Q(v)$ and $\left(v_1^T v_1\right)\left(v_2^T v_2\right) = v_1^T (v_1 \cdot v_2)\, v_2$ has trace $(v_1 \cdot v_2)^2$,

$$\operatorname{tr}\left(\sigma_{v_1}\sigma_{v_2}\right) = 3 - 2 - 2 + \frac{4(v_1 \cdot v_2)^2}{Q(v_1)\, Q(v_2)} = -1 + 4(1 - s(v_1, v_2))$$

so $p = \frac{3 - \operatorname{tr}}{4} = s(v_1, v_2)$. ∎

Classically, two reflections rotate through twice the angle between the mirrors. Algebraically, the spread between the normals is exactly the half spread of the product. In particular, perpendicular normals give a half turn.

Example 36 *For $v_1 = (1,2,3)$ and $v_2 = (2,1,-1)$ of Section 4, $s(v_1,v_2) = 1 - \frac{1}{84} = \frac{83}{84}$, and indeed the product computed there has trace $\frac{-8+4-16}{21} = -\frac{20}{21}$, giving $p = \frac{1}{4}\left(3 + \frac{20}{21}\right) = \frac{83}{84}$.*

7 The anchor of a rotation

Definition 37 *The **big Cayley transform** of a matrix z is*

$$C(z) \equiv (I - z)(I + z)^{-1}$$

*whenever $I + z$ is invertible, and the **anchor** of a rotation ρ with $p(\rho) \neq 1$ is*

$$A(\rho) \equiv -C(\rho) = (\rho - I)(\rho + I)^{-1}.$$

By the characteristic polynomial of Section 6, -1 is an eigenvalue of ρ precisely when the quadratic factor vanishes at $X = -1$, that is when $4(1-p) = 0$. So $\det(I + \rho) \neq 0$ precisely when $p(\rho) \neq 1$, and the anchor is defined exactly there.

Theorem 38 (Anchor of a rotation) *For a rotation ρ with $p \equiv p(\rho) \neq 1$:*

1. $A(\rho)$ is antisymmetric, so $A(\rho) = A_w$ for a unique vector w;

2. $c(A(\rho)) = \rho$ and $A(c(A)) = A$: the cap and the anchor are mutually inverse bijections between $\{A_w : Q(w) \neq -1\}$ and $\{\rho : p(\rho) \neq 1\}$;

3. $w\rho = w$: the rotation fixes its anchor vector;

4. $Q(w) = p/(1-p)$, the half twist of ρ;

5. writing $\rho^{as} \equiv \frac{1}{2}\left(\rho - \rho^T\right)$ for the antisymmetric part,

$$A(\rho) = \frac{2}{1 + \operatorname{tr}\rho}\rho^{as}.$$

Proof. For 1: since $\rho^T = \rho^{-1}$,

$$A(\rho)^T = \left(\rho^{-1} - I\right)\left(\rho^{-1} + I\right)^{-1} = (I - \rho)\rho^{-1} \cdot \rho(I + \rho)^{-1} = -A(\rho).$$

For 2: the maps $w \mapsto (I + w)(I - w)^{-1}$ and $z \mapsto (z - I)(z + I)^{-1}$ are mutually inverse wherever defined, all factors in sight being commuting polynomials in a single matrix; part 4 of the theorem that caps are rotations shows the domains match. For 5: write $\rho = c(A_w) = I + \frac{2}{1+Q}(A_w + A_w^2)$ with $Q = Q(w)$; the term $A_w^2 = w^T w - QI$ is symmetric, so

$$\rho^{as} = \frac{2}{1+Q}A_w, \qquad 1 + \operatorname{tr}\rho = 1 + \frac{3-Q}{1+Q} = \frac{4}{1+Q},$$

and dividing gives the formula. Then 3 follows from part 2 of the theorem that caps are rotations, and 4 from the half spread identities. ∎

Part 5 is computationally useful. For a rotation which is not a half turn, the anchor is obtained by antisymmetrization and one scalar division. No matrix inversion is required. The cap then recovers the rotation, so the anchor is a complete invariant on this affine chart.

Example 39 *For the rotation* $\rho = \frac{1}{67}\begin{pmatrix} -33 & 22 & 54 \\ -6 & -63 & 22 \\ 58 & 6 & 33 \end{pmatrix}$ *of Section 5,* $\operatorname{tr}\rho = -\frac{63}{67}$ *and* $1+\operatorname{tr}\rho = \frac{4}{67}$*, while*

$$\rho^{as} = \frac{1}{67}\begin{pmatrix} 0 & 14 & -2 \\ -14 & 0 & 8 \\ 2 & -8 & 0 \end{pmatrix}, \qquad A(\rho) = \frac{67}{2}\rho^{as} = \begin{pmatrix} 0 & 7 & -1 \\ -7 & 0 & 4 \\ 1 & -4 & 0 \end{pmatrix} = A_{(4,1,7)}$$

recovering the anchor vector $(4,1,7)$ *exactly.*

7.1 The anchor of a product of two reflections

Theorem 40 (Anchor of two reflections) *If* v_1 *and* v_2 *are non-null with* $d \equiv v_1 \cdot v_2 \neq 0$*, then*

$$A(\sigma_{v_1}\sigma_{v_2}) = \frac{v_1^T v_2 - v_2^T v_1}{v_1 \cdot v_2} = \frac{1}{d}A_{v_1 \times v_2}$$

which depends only on the lines $[v_1]$ *and* $[v_2]$*, and has quadrance* $s/(1-s)$ *where* $s = s(v_1, v_2)$*.*

Proof. Write $Q_1 \equiv Q(v_1)$ and $Q_2 \equiv Q(v_2)$. Expanding $\rho \equiv \sigma_{v_1}\sigma_{v_2}$, the terms $v_1^T v_1$ and $v_2^T v_2$ are symmetric, and the remaining term is $\frac{4d}{Q_1 Q_2}v_1^T v_2$, so by part 2 of the corollary of Section 3

$$\rho^{as} = \frac{2d}{Q_1 Q_2}\left(v_1^T v_2 - v_2^T v_1\right) = \frac{2d}{Q_1 Q_2}A_{v_1 \times v_2},$$

while by the spread of normals theorem $1+\operatorname{tr}\rho = 4(1-s) = \frac{4d^2}{Q_1 Q_2}$. The formula follows from part 5 of the anchor of a rotation theorem, the line dependence is visible since scaling v_1 or v_2 scales numerator and denominator equally, and the quadrance is

$$\frac{Q(v_1 \times v_2)}{d^2} = \frac{Q_1 Q_2 - d^2}{d^2} = \frac{s}{1-s}$$

by the Lagrange identity. ∎

The excluded case $d = 0$ is exactly right: perpendicular mirrors compose to a half turn ($s = 1$, $p = 1$), which has no anchor.

Example 41 *For* $v_1 = (1,2,3)$ *and* $v_2 = (2,1,-1)$ *of Sections 4 and 6,* $d = 1$ *and* $A(\sigma_{v_1}\sigma_{v_2}) = A_{(-5,7,-3)}$ *with quadrance* $25+49+9 = 83 = \frac{83/84}{1/84}$*, confirming the half twist.*

7.2 Proof of the two reflections theorem

We now prove the two reflections theorem of Section 4, in the following sharper form: if $\rho \neq I$, the first mirror line may be chosen freely among the non-null lines perpendicular to the anchor vector of ρ, or to the axis of ρ when ρ is a half turn, and the second mirror line is then determined.

Proof. The identity is $\sigma_v\sigma_v$ for any non-null v. In every case the second mirror is determined by the first, since $\sigma_{v_1}\sigma_{v_2} = \rho$ forces $\sigma_{v_2} = \sigma_{v_1}\rho$, and a reflection determines its line by part 2 of the reflections theorem.

If ρ is a half turn $\rho_{[v]}$, choose any non-null v_1 in the plane $v^\perp$, and let v_2 span the line of $v^\perp$ perpendicular to v_1. The restriction of the dot product to $v^\perp$ is non-degenerate, so v_2 is non-null: otherwise it would be perpendicular to all of $v^\perp$. Now $\sigma_{v_1}\sigma_{v_2}$ negates v_1 and v_2, hence every vector of $v^\perp$, and fixes v, which is perpendicular to both mirrors; by the uniqueness in the half turns theorem it equals $\rho_{[v]}$.

Otherwise $\rho = c(A_w)$ with $Q \equiv Q(w) \neq -1$ and $w \neq 0$. Choose any non-null v_1 with $v_1 \cdot w = 0$, and set

$$v_2 \equiv v_1(I + A_w) = v_1 + w \times v_1.$$

Then $v_1 \cdot v_2 = Q(v_1) \neq 0$ since $w \times v_1 \perp v_1$; and $Q(v_2) = Q(v_1) + Q(w \times v_1) = Q(v_1)(1+Q) \neq 0$ by the Lagrange identity, so v_2 is non-null; and

$$v_1 \times v_2 = v_1 \times (w \times v_1) = wQ(v_1) - v_1(w \cdot v_1) = Q(v_1)\, w$$

by the vector triple product identity. Hence by the anchor of two reflections theorem $A(\sigma_{v_1}\sigma_{v_2}) = \frac{1}{Q(v_1)} A_{Q(v_1)w} = A_w$, and since the anchor is a complete invariant, $\sigma_{v_1}\sigma_{v_2} = \rho$. ■

8 Orientation and handedness

A rotation and its inverse have the same axis and the same half spread. Their anchors have opposite signs:

$$A\left(\rho^{-1}\right) = A\left(\rho^T\right) = -A(\rho).$$

Thus the anchor also records a sense of rotation along the axis. We now express this by a determinant.

Theorem 42 (Orientation determinant) *Let $\rho = c(A_v)$ with $Q \equiv Q(v) \neq -1$, and let $w \perp v$. Then*

$$\det \begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} = 2p(1-p)(1+Q)\,Q(w) = \frac{2Q}{1+Q} Q(w),$$

where $p = p(\rho)$.

Proof. For $w \perp v$, the spread calculation gives

$$w\rho = \frac{(1-Q)\,w + 2(v \times w)}{1+Q}.$$

The first term contributes zero to the determinant, so

$$\det \begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} = \frac{2}{1+Q} \det \begin{pmatrix} v \\ w \\ v \times w \end{pmatrix} = \frac{2}{1+Q} Q(v \times w).$$

Since $v \cdot w = 0$, the Lagrange identity gives $Q(v \times w) = QQ(w)$. The half spread identities give the other form. ■

Corollary 43 (Orientation test) *Over an ordered field, if $v \neq 0$, $w \neq 0$ and $w \perp v$, then*

$$\det \begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} > 0 \qquad \textit{for} \qquad \rho = c\left(A_v\right).$$

With the first row v kept fixed, the inverse rotation $c\left(A_{-v}\right)$ gives a negative determinant. Thus the two anchor vectors on a non-null axis distinguish a rotation from its inverse.

Proof. Over an ordered field $Q\left(v\right) > 0$ and $Q\left(w\right) > 0$, so the determinant is positive. The theorem applied to the anchor vector $-v$ shows that the frame $-v, w, wc\left(A_{-v}\right)$ has positive determinant, so the frame $v, w, wc\left(A_{-v}\right)$ has negative determinant. ■

The orientation convention is already present in our chosen cross product, and hence in the sign used to define A_v. The theorem does not choose that convention. Rather, once the convention is fixed, it replaces the usual right-hand mnemonic by an algebraic determinant test.

Example 44 *For $v = (4, 1, 7)$ and $w = (1, 3, -1)$ we have $Q\left(v\right) = 66$, $Q\left(w\right) = 11$ and $v \cdot w = 0$. Therefore*

$$\det \begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} = \frac{2 \cdot 66}{67} \cdot 11 = \frac{1452}{67} > 0.$$

For the inverse rotation $c\left(A_{-v}\right)$, with the same first row v, the determinant has the opposite sign.

Example 45 *Over $\mathbb{F}_5$ let $v = (1, 2, 0)$ be null and $\rho = c\left(A_v\right)$. Then the determinant above is 0 for every $w \perp v$, although $c\left(A_v\right)$ and $c\left(A_{-v}\right)$ are distinct inverse parabolic rotations. A null axis therefore behaves differently: the axis and half spread do not reduce the possibilities to one inverse pair. The full anchor still records the missing parameter.*

Over an anisotropic ordered field such as $\mathbb{Q}$, a non-identity, non-half-turn rotation is therefore determined by its axis, its half spread and its orientation choice. Over fields with null axes, the anchor is the safer invariant.

9 The anchor product

Rotations compose; we now compute the composition on anchors. Rotations act on row vectors from the right, so $\rho_1\rho_2$ performs ρ_1 first.

Definition 46 *For vectors v and u with $v \cdot u \neq 1$, the **anchor product** is*

$$v \otimes u \equiv \frac{v + u - v \times u}{1 - v \cdot u}$$

and correspondingly for anchors, using the anchor form of Section 3,

$$A \otimes B \equiv \frac{A + B + (AB - BA)}{1 - A \cdot B}$$

so that $A_v \otimes A_u = A_{v \otimes u}$, by part 2 of the corollary of Section 3.

Lemma 47 (Quadrance of an anchor product) *For any vectors v and u,*

$$(1 - v \cdot u)^2 + Q(v + u - v \times u) = (1 + Q(v))(1 + Q(u)).$$

Consequently if $v \cdot u \neq 1$ then $1 + Q(v \otimes u) = \frac{(1+Q(v))(1+Q(u))}{(1-v\cdot u)^2}$.

Proof. Write $d \equiv v \cdot u$. Since $v \times u$ is perpendicular to v and u,

$$Q(v + u - v \times u) = Q(v) + Q(u) + 2d + Q(v \times u) = Q(v) + Q(u) + 2d + Q(v)\,Q(u) - d^2$$

by the Lagrange identity, and adding $(1 - d)^2 = 1 - 2d + d^2$ gives $1 + Q(v) + Q(u) + Q(v)\,Q(u)$. ■

So if the caps of A_v and A_u are defined and $v \cdot u \neq 1$, the cap of $A_{v\otimes u}$ is also defined.

Theorem 48 (Composition) *If $Q(v) \neq -1$, $Q(u) \neq -1$ and $v \cdot u \neq 1$, then*

$$c(A_v)\,c(A_u) = c(A_{v\otimes u}).$$

Proof. Write $\rho_1 \equiv c(A_v)$, $\rho_2 \equiv c(A_u)$, $d \equiv v \cdot u$ and $m \equiv v \times u$. We first treat the case where m is non-null, using reflections, and then use the vanishing lemma for the general case.

Suppose $Q(m) \neq 0$, and set

$$a \equiv m(I - A_v) = m - v \times m, \qquad b \equiv m(I + A_u) = m + u \times m.$$

Since $m \perp v$ and $m \perp u$, the construction of Section 7.2, applied to $\rho_1^{-1} = c(A_{-v})$ with first mirror m, gives $\sigma_m\sigma_a = \rho_1^{-1}$, that is $\rho_1 = \sigma_a\sigma_m$; applied to ρ_2 with first mirror m it gives $\rho_2 = \sigma_m\sigma_b$. (The mirrors are non-null: by the Lagrange identity $Q(a) = Q(m)(1 + Q(v))$ and $Q(b) = Q(m)(1 + Q(u))$, all factors non-zero.) The shared middle mirror now cancels:

$$\rho_1\rho_2 = \sigma_a\sigma_m\sigma_m\sigma_b = \sigma_a\sigma_b.$$

By Binet–Cauchy, using $v \cdot m = u \cdot m = 0$,

$$a \cdot b = Q(m) - (v \times m) \cdot (u \times m) = Q(m) - dQ(m) = (1 - d)\,Q(m) \neq 0,$$

so the anchor of two reflections theorem applies. By the vector triple product identity, $m \times (u \times m) = uQ(m)$ and $m \times (v \times m) = vQ(m)$, while

$$(v \times m) \times (u \times m) = -(v \cdot (m \times u))\,m = Q(m)\,m$$

since $(m \times u) \cdot v = \det(m, u, v) = (u \times v) \cdot m = -Q(m)$. Hence

$$a \times b = vQ(m) + uQ(m) - Q(m)\,m = Q(m)(v + u - v \times u)$$

and therefore

$$A(\rho_1\rho_2) = \frac{1}{a \cdot b} A_{a\times b} = A_{\frac{v+u-v\times u}{1-d}} = A_{v\otimes u}.$$

Since the anchor is a complete invariant, $\rho_1\rho_2 = c(A_{v\otimes u})$.

For the general case write $n \equiv v + u - v \times u$, so that $v \otimes u = \frac{1}{1-d} n$. By quadratic reduction and the lemma,

$$(1 + Q(v))(1 + Q(u))\, c(A_{v \otimes u}) = (1 + Q(v))(1 + Q(u))\, I + 2\left((1 - d) A_n + A_n^2\right),$$

while

$$(1 + Q(v))(1 + Q(u))\, c(A_v)\, c(A_u) = \left((1 + Q(v))\, I + 2\left(A_v + A_v^2\right)\right)\left((1 + Q(u))\, I + 2\left(A_u + A_u^2\right)\right).$$

The difference P of the two right hand sides is a matrix of polynomials with integer coefficients in the six coordinates of v and u, and the theorem asserts precisely that $P = 0$. Over $\mathbb{Q}$, where $1 + Q(v)$ and $1 + Q(u)$ never vanish, the first part of the proof shows that P vanishes at every rational point where $h \equiv Q(v \times u)(1 - v \cdot u)$ does not vanish, and h is not the zero polynomial. By the vanishing lemma of Section 2, P is the zero polynomial, so $P = 0$ over every field. ■

Example 49 *Let* $v = (4, 1, 5)$ *and* $u = (-3, 4, 2)$*, so* $d = 2$ *and* $v \times u = (-18, -23, 19)$*. Then*

$$v \otimes u = \frac{(1, 5, 7) - (-18, -23, 19)}{1 - 2} = (-19, -28, 12)$$

so the product of the rotations with anchor vectors $(4, 1, 5)$ *and* $(-3, 4, 2)$*, of half spreads* $\frac{42}{43}$ *and* $\frac{29}{30}$*, is the rotation with anchor vector* $(-19, -28, 12)$*, of half spread* $\frac{1289}{1290}$*—one line of vector arithmetic, in place of a product of two* 3×3 *matrices followed by an axis extraction. As the lemma predicts,* $1 + Q(-19, -28, 12) = 1290 = 43 \cdot 30$*.*

Corollary 50 (Half turn criterion) *For* $Q(v), Q(u) \neq -1$*, the product* $c(A_v)\, c(A_u)$ *is a half turn exactly when* $v \cdot u = 1$*, and in that case it is the half turn about* $[v + u - v \times u]$*.*

Proof. If $v \cdot u \neq 1$ the product is a cap by the composition theorem, so -1 is not an eigenvalue, while every half turn negates a non-zero plane; so the product is not a half turn. If $d \equiv v \cdot u = 1$, write $n \equiv v + u - v \times u$. The quadrance of an anchor product lemma gives $Q(n) = (1 + Q(v))(1 + Q(u)) \neq 0$, and the polynomial identity $P = 0$ from the proof of the composition theorem, divided by this quantity, reads

$$c(A_v)\, c(A_u) = I + \frac{2A_n^2}{Q(n)} = -I + \frac{2}{Q(n)} n^T n = \rho_{[n]},$$

using $A_n^2 = n^T n - Q(n)\, I$. ■

Corollary 51 (Tangent addition) *For a fixed non-zero vector* a *with* $Q \equiv Q(a)$*, the caps* $c(xA_a)$ *for* x *in* $\mathbb{F}$ *compose by*

$$c(xA_a)\, c(yA_a) = c(zA_a), \qquad z = \frac{x + y}{1 - Qxy},$$

whenever the caps on the left are defined and $Qxy \neq 1$*: the one-parameter family through a given rotation composes by the rational addition law with modulus* Q*.*

Proof. For $v = xa$ and $u = ya$ we have $v \times u = 0$ and $v \cdot u = Qxy$, so $v \otimes u = \frac{x+y}{1-Qxy} a$. ■

Example 52 *Take $a = (1,0,0)$, so $Q = 1$. By the coordinate form of the cap, $c(xA_a)$ fixes a and acts on the last two coordinates by the block $\frac{1}{1+x^2}\begin{pmatrix} 1-x^2 & 2x \\ -2x & 1-x^2 \end{pmatrix}$. For $x = \frac{1}{2}$ this is $\begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{pmatrix}$, the 3–4–5 rotation, which reappears in Section 12. Its square has $z = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$ and block $\begin{pmatrix} -\frac{7}{25} & \frac{24}{25} \\ -\frac{24}{25} & -\frac{7}{25} \end{pmatrix}$, the 7–24–25 rotation. The half spreads are $\frac{1}{5}$ and $\frac{16}{25} = 4 \cdot \frac{1}{5} \cdot \frac{4}{5}$, as the half spread of a square theorem requires.*

Remark 53 (Opposite product) *The companion operation $v \otimes^{op} u \equiv \frac{v+u+v\times u}{1-v\cdot u}$ satisfies $v \otimes^{op} u = u \otimes v$, so by the composition theorem it corresponds to composing the rotations in the opposite order: $c(A_u)\, c(A_v) = c(A_{v \otimes^{op} u})$.*

Theorem 54 (Conditional associativity) *The anchor product is associative wherever all products occurring are defined: if $v \otimes u$, $(v \otimes u) \otimes w$, $u \otimes w$ and $v \otimes (u \otimes w)$ all exist, and the caps of A_v, A_u and A_w are defined, then*

$$(v \otimes u) \otimes w = v \otimes (u \otimes w).$$

It is genuinely partial: one side may exist while the other does not.

Proof. When all four products exist, both sides are anchor vectors of the single rotation $c(A_v)\, c(A_u)\, c(A_w)$ by the composition theorem, and anchors are unique. For the partiality, take v and u with $v \cdot u = 1$ and $Q(u) \neq -1$, and $w = -u$: then $u \otimes w = 0$ and $v \otimes (u \otimes w) = v$ exists, while $v \otimes u$ does not. ∎

The failure of totality is not a blemish to be excused but a signpost: the anchor product is the group law of $SO(3)$ seen through a chart that misses the half turns, and every defect of the product is the shadow of that missing locus.

10 Four breakdowns of the affine theory

The theory so far parametrizes rotations by anchors: a rotation is $c(A_v)$, computations on rotations become vector arithmetic on the anchor vectors v, and the dictionary is complete except at four points, which we now exhibit in their sharpest forms. Over $\mathbb{Q}$ only the first three are visible.

First breakdown: half turns have no anchors. The cap misses exactly the rotations with $p = 1$ (Sections 5 and 7). These missing rotations form a family parametrized by the non-null lines of $\mathbb{F}^3$—a projective plane, with the null conic removed when the field has null vectors.

Second breakdown: the anchor product has a forbidden locus. The product $v \otimes u$ fails exactly when $v \cdot u = 1$, and by the half turn criterion of Section 9 this is no accident of the formula: the composed rotation *is* a half turn, so there is no anchor for the product to be.

Example 55 *Let $v_1 = (4,1,7)$ and $v_4 = (-2,2,1)$, so that $v_1 \cdot v_4 = -8+2+7 = 1$. The rotations $\rho_1 = c(A_{v_1})$ and $\rho_4 = c(A_{v_4})$ are unexceptional, with half spreads $\frac{66}{67}$ and $\frac{9}{10}$, yet their product*

$$\rho_5 \equiv \rho_1\rho_4 = \begin{pmatrix} -\frac{22}{67} & \frac{63}{67} & -\frac{6}{67} \\ \frac{63}{67} & \frac{106}{335} & -\frac{42}{335} \\ -\frac{6}{67} & -\frac{42}{335} & -\frac{331}{335} \end{pmatrix}$$

has characteristic polynomial $(X-1)(X+1)^2$*: a half turn. The numerator of the anchor product nevertheless survives the death of its denominator, and delivers the axis:*

$$v_1 + v_4 - v_1 \times v_4 = (2,3,8) - (-13,-18,10) = (15,21,-2)$$

and indeed $\rho_5 = \rho_{[(15,21,-2)]}$*, the half turn about* $(15,21,-2)$*.*

The same example shows the projective repair. Introduce a parameter and consider the anchors $\frac{1}{x}A_{(15,21,-2)}$ for $x \neq 0$: by quadratic reduction, with $Q = Q(15,21,-2) = 670$,

$$f(x) \equiv c\left(\tfrac{1}{x}A\right) = I + \frac{2(xA + A^2)}{x^2 + Q}, \qquad A \equiv A_{(15,21,-2)},$$

and the right side is defined at $x = 0$:

$$f(0) = I + \frac{2}{670}A^2 = -I + \frac{2}{670}(15,21,-2)^T(15,21,-2) = \rho_5.$$

No limit is taken and none is needed: the substitution $x = 0$ is field arithmetic. What the affine chart parametrizes by the anchor vector $\frac{1}{x}(15,21,-2)$, which loses meaning at $x = 0$, the pair $(x : (15,21,-2))$ parametrizes uniformly—the half turn is the point where the *proportion* is $(0 : (15,21,-2))$, an **anchor at infinity**.

Third breakdown: associativity is conditional. By Section 9 the anchor product is associative only where defined, and a triple product can exist with one bracketing and not the other. The group $SO(3)$ is associative; the defect belongs entirely to the parametrization.

Fourth breakdown: an excluded quadric of anchors. Over a field in which $Q(v) = -1$ is solvable—every finite field of odd characteristic, since every element there is a sum of three squares—the anchors A_v with $Q(v) = -1$ have no cap (Section 5) and correspond to no rotation at all. Over $\mathbb{Q}$, and over any ordered field, this quadric is empty, which is why only three breakdowns are visible rationally; in the relativistic theory of [17] it is rational and unavoidable.

The four breakdowns share a single cause and a single cure. The anchor vector v is the affine chart $x = 1$ of a projective object, the proportion $(x : v) = (x : t : r : s)$; the half turns are its points at infinity; the forbidden locus $v \cdot u = 1$ is where a product lands at infinity; conditional associativity is what associativity looks like through a chart that cannot see infinity; and the excluded quadric is the affine trace, at $x = 1$, of the quadric of proportions with $x^2 + Q(v) = 0$. In the next section we abandon the chart.

11 Quaternions and the projective rotation group

Definition 56 *A* ***proportion*** *is a non-zero vector* $(x,v) = (x,t,r,s) \in \mathbb{F}^4$ *taken up to multiplication by non-zero scalars, written* $(x : v) = (x : t : r : s)$*; the space of proportions is the projective space* $P^3(\mathbb{F})$*. The* ***quadrance*** *of a proportion is the square class of*

$$N(x,v) \equiv x^2 + Q(v) = x^2 + t^2 + r^2 + s^2,$$

and the proportion is ***null*** *exactly when* $N(x,v) = 0$*.*

Over $\mathbb{Q}$, or any ordered field, there are no null proportions.

Definition 57 *The **rotation of a non-null proportion** is*

$$\rho(x:v) \equiv I + \frac{2(xA_v + A_v^2)}{x^2 + Q(v)}.$$

This is well defined: replacing (x, v) by $(\lambda x, \lambda v)$ multiplies numerator and denominator by λ^2. On the chart $x \neq 0$, normalizing to $x = 1$, it is the cap: $\rho(1:v) = c(A_v)$ by quadratic reduction. At infinity it is the half turn: for v non-null

$$\rho(0:v) = I + \frac{2A_v^2}{Q(v)} = -I + \frac{2}{Q(v)} v^T v = \rho_{[v]}.$$

Moreover, by the quadrance of an anchor product lemma, the polynomial identity $P = 0$ established in the proof of the composition theorem reads

$$c(A_v)\, c(A_u) = \rho(1 - v \cdot u : v + u - v \times u)$$

for all v, u with $Q(v), Q(u) \neq -1$, with no restriction on $v \cdot u$: when $v \cdot u = 1$ the right side is the half turn about $v + u - v \times u$, exactly as in the example of Section 10.

Theorem 58 (Half spread of a proportion) *For a non-null proportion $(x:v)$ with $N \equiv N(x, v)$,*

$$p(\rho(x:v)) = \frac{Q(v)}{N}, \qquad 1 - p(\rho(x:v)) = \frac{x^2}{N},$$

and $\rho(x:-v) = \rho(x:v)^T = \rho(x:v)^{-1}$.

Proof. Since $\operatorname{tr} A_v = 0$ and $\operatorname{tr} A_v^2 = -2Q(v)$ by the Isometry theorem, $\operatorname{tr} \rho(x:v) = 3 - \frac{4Q(v)}{N}$, so $p = \frac{Q(v)}{N}$ and $1 - p = \frac{N - Q(v)}{N} = \frac{x^2}{N}$. Replacing v by $-v$ replaces A_v by $A_{-v} = -A_v = A_v^T$ and leaves A_v^2 and N unchanged, which transposes $\rho(x:v)$; and the transpose of a rotation is its inverse. ∎

Thus the scalar x and the vector v of a proportion divide the quadrance N between them, x^2 carrying the part $1 - p$ and $Q(v)$ the part p; and negating the vector part inverts the rotation, as negating the anchor did in Section 8.

11.1 The quaternion algebra

The projective product comes from a four dimensional associative algebra.

Definition 59 *A **quaternion** over $\mathbb{F}$ is a vector $\alpha = (x, v) = (x, t, r, s) \in \mathbb{F}^4$, with **scalar part** x and **vector part** v; it is **pure** when $x = 0$, and we identify a vector $w \in \mathbb{F}^3$ with the pure quaternion $(0, w)$ and a scalar $x \in \mathbb{F}$ with $(x, 0)$. The **quaternion product** of $\alpha = (x, v)$ and $\beta = (y, u)$ is*

$$\alpha \otimes \beta \equiv (xy - v \cdot u,\ xu + yv - v \times u),$$

*the **conjugate** of α is $\overline{\alpha} \equiv (x, -v)$, and the **quadrance** of α is $N(\alpha) \equiv x^2 + Q(v)$. The quaternion α is **null** exactly when $N(\alpha) = 0$.*

In coordinates, with $\alpha = (x_1, t_1, r_1, s_1)$ and $\beta = (x_2, t_2, r_2, s_2)$, the product $\alpha \otimes \beta = (x, t, r, s)$ has

$$\begin{aligned} x &= x_1 x_2 - t_1 t_2 - r_1 r_2 - s_1 s_2 \\ t &= x_1 t_2 + x_2 t_1 - r_1 s_2 + r_2 s_1 \\ r &= x_1 r_2 + x_2 r_1 - s_1 t_2 + s_2 t_1 \\ s &= x_1 s_2 + x_2 s_1 - t_1 r_2 + t_2 r_1. \end{aligned}$$

Writing $\mathbf{i} \equiv (0,1,0,0)$, $\mathbf{j} \equiv (0,0,1,0)$, $\mathbf{k} \equiv (0,0,0,1)$ for the pure standard basis quaternions, so that $(x,t,r,s) = x + t\mathbf{i} + r\mathbf{j} + s\mathbf{k}$, the product is determined by bilinearity and the table

$$\mathbf{i} \otimes \mathbf{i} = \mathbf{j} \otimes \mathbf{j} = \mathbf{k} \otimes \mathbf{k} = -1, \qquad \mathbf{j} \otimes \mathbf{i} = \mathbf{k}, \quad \mathbf{k} \otimes \mathbf{j} = \mathbf{i}, \quad \mathbf{i} \otimes \mathbf{k} = \mathbf{j},$$

with $\mathbf{i} \otimes \mathbf{j} = -\mathbf{k}$, $\mathbf{j} \otimes \mathbf{k} = -\mathbf{i}$ and $\mathbf{k} \otimes \mathbf{i} = -\mathbf{j}$.

Theorem 60 (Quaternion algebra) *For quaternions α, β, γ over $\mathbb{F}$:*

1. *the product is bilinear, with identity $1 = (1,0)$, and associative: $(\alpha \otimes \beta) \otimes \gamma = \alpha \otimes (\beta \otimes \gamma)$;*
2. *$\overline{\alpha \otimes \beta} = \overline{\beta} \otimes \overline{\alpha}$;*
3. *$\alpha \otimes \overline{\alpha} = \overline{\alpha} \otimes \alpha = N(\alpha)$;*
4. *$N(\alpha \otimes \beta) = N(\alpha) N(\beta)$;*
5. *a non-null quaternion is invertible, with $\alpha^{-1} = \overline{\alpha}/N(\alpha)$, and a null quaternion is not.*

Proof. Bilinearity and the identity are visible. For associativity, let $\alpha = (x, v)$, $\beta = (y, u)$, $\gamma = (z, w)$. The scalar part of $(\alpha \otimes \beta) \otimes \gamma$ is

$$(xy - v \cdot u) z - (xu + yv - v \times u) \cdot w = xyz - z(v \cdot u) - x(u \cdot w) - y(v \cdot w) + (v \times u) \cdot w$$

and that of $\alpha \otimes (\beta \otimes \gamma)$ is

$$x(yz - u \cdot w) - v \cdot (yw + zu - u \times w) = xyz - x(u \cdot w) - y(v \cdot w) - z(v \cdot u) + v \cdot (u \times w),$$

which agree by the scalar triple product identity $(v \times u) \cdot w = v \cdot (u \times w)$. The vector part of $(\alpha \otimes \beta) \otimes \gamma$ is

$$xyw + xzu + yzv - z(v \times u) - x(u \times w) - y(v \times w) - (v \cdot u) w + (v \times u) \times w$$

and that of $\alpha \otimes (\beta \otimes \gamma)$ is

$$xyw + xzu + yzv - z(v \times u) - x(u \times w) - y(v \times w) - (u \cdot w) v + v \times (u \times w).$$

The first six terms coincide, and by the vector triple product identity $(v \times u) \times w = u(v \cdot w) - v(u \cdot w)$ while $v \times (u \times w) = u(v \cdot w) - w(v \cdot u)$, so the remaining two terms on each side sum to $u(v \cdot w) - v(u \cdot w) - w(v \cdot u)$ on both. For 2,

$$\overline{\beta} \otimes \overline{\alpha} = (y, -u) \otimes (x, -v) = (xy - u \cdot v,\ -yv - xu - u \times v) = (xy - v \cdot u,\ -(xu + yv - v \times u)) = \overline{\alpha \otimes \beta}.$$

For 3, $(x, v) \otimes (x, -v) = (x^2 + v \cdot v,\ -xv + xv + v \times v) = (N(\alpha), 0)$, and the other order likewise. For 4, by 1, 2 and 3,

$$N(\alpha \otimes \beta) = (\alpha \otimes \beta) \otimes \overline{\alpha \otimes \beta} = \alpha \otimes \left(\beta \otimes \overline{\beta}\right) \otimes \overline{\alpha} = N(\beta)\ \alpha \otimes \overline{\alpha} = N(\alpha) N(\beta).$$

For 5, if $N(\alpha) \neq 0$ then 3 exhibits the inverse; if $N(\alpha) = 0$ and $\alpha \otimes \beta = 1$ then 4 gives $0 = 1$. ∎

Remark 61 (Hamilton's convention) *Hamilton's product is ours in the opposite order: $\alpha\beta = \beta \otimes \alpha$, so that Hamilton has $\mathbf{ij} = \mathbf{k}$ where we have $\mathbf{i} \otimes \mathbf{j} = -\mathbf{k}$. The reversal is forced by our use of row vectors acted on from the right, under which $\rho_1\rho_2$ means ρ_1 first; the theorem below then reads $\rho(\alpha \otimes \beta) = \rho(\alpha)\,\rho(\beta)$ with no reversal. Nothing is lost: by part 2 of the theorem, conjugation $\alpha \mapsto \overline{\alpha}$ is an isomorphism from Hamilton's algebra onto ours, and the opposite product $\otimes^{op}$ of Section 9 is Hamilton's.*

Corollary 62 (Euler's four square identity) *For any eight elements of $\mathbb{F}$,*

$$\begin{aligned}&\left(x_1^2 + t_1^2 + r_1^2 + s_1^2\right)\left(x_2^2 + t_2^2 + r_2^2 + s_2^2\right)\\ &= \left(x_1x_2 - t_1t_2 - r_1r_2 - s_1s_2\right)^2 + \left(x_1t_2 + x_2t_1 - r_1s_2 + r_2s_1\right)^2\\ &\qquad + \left(x_1r_2 + x_2r_1 - s_1t_2 + s_2t_1\right)^2 + \left(x_1s_2 + x_2s_1 - t_1r_2 + t_2r_1\right)^2.\end{aligned}$$

Proof. This is part 4 of the theorem written out in coordinates. ■

Euler found this identity in 1748 [3], and it is the multiplicative step in Lagrange's theorem that every positive integer is a sum of four squares: once every prime is such a sum, the identity extends the property to all products. We use Lagrange's theorem in Section 12; here we note that the geometry of rotations returns the identity as a by-product. The quadrance of an anchor product lemma of Section 9 is the special case $x_1 = x_2 = 1$.

11.2 The projective rotation theorem

Definition 63 *The **quaternion product of proportions** is*

$$(x : v) \otimes (y : u) \equiv (xy - v \cdot u \;:\; xu + yv - v \times u),$$

that is, the proportion of the quaternion product of any representatives.

This is well defined since the quaternion product is bilinear: rescaling the representatives rescales the product.

Theorem 64 (The product is total) *If $(x : v)$ and $(y : u)$ are non-null proportions then $(x : v) \otimes (y : u)$ is a non-null proportion, with quadrance $N(x, v)\,N(y, u)$, and the product of proportions is associative. The non-null proportions form a group under $\otimes$, with identity $(1 : 0)$ and inverse $(x : v)^{-1} = (x : -v)$.*

Proof. By parts 4, 1, 3 and 5 of the quaternion algebra theorem, since $(x : -v)$ is the proportion of $\overline{\alpha}$ and $\alpha \otimes \overline{\alpha} = N(\alpha)$ is a non-zero scalar, whose proportion is $(1 : 0)$. ■

Theorem 65 (Sandwich formula) *For a non-null proportion $(x : v)$ with representative $\alpha = (x, v)$ and any vector w,*

$$w\,\rho(x : v) = \frac{\overline{\alpha} \otimes w \otimes \alpha}{N(\alpha)}$$

where w is identified with the pure quaternion $(0, w)$; the right side is again pure.

Proof. First $\overline{\alpha} \otimes w = (x, -v) \otimes (0, w) = (v \cdot w,\ xw + v \times w)$. Then

$$(v \cdot w,\ xw + v \times w) \otimes (x, v) = (x\,(v \cdot w) - (xw + v \times w) \cdot v,\ (v \cdot w)\,v + x\,(xw + v \times w) - (xw + v \times w) \times v)\,.$$

The scalar part is $x\,(v \cdot w) - x\,(w \cdot v) - 0 = 0$. In the vector part, $-(xw) \times v = x\,(v \times w)$ and $-(v \times w) \times v = v \times (v \times w) = v\,(v \cdot w) - wQ\,(v)$ by the vector triple product identity, so the vector part is

$$\left(x^2 - Q\,(v)\right) w + 2x\,(v \times w) + 2\,(v \cdot w)\,v.$$

On the other side, by the cross product lemma and part 1 of the corollary of Section 3, $wA_v = v \times w$ and $wA_v^2 = (w \cdot v)\,v - Q\,(v)\,w$, so

$$N\,(\alpha)\ w\rho\,(x : v) = \left(x^2 + Q\,(v)\right) w + 2x\,(v \times w) + 2\,(w \cdot v)\,v - 2Q\,(v)\,w,$$

which is the same. ■

Theorem 66 (Projective rotation theorem) *The map $(x : v) \mapsto \rho\,(x : v)$ is a bijection from the non-null proportions of $P^3\,(\mathbb{F})$ onto $SO\,(3, \mathbb{F})$, and it is multiplicative:*

$$\rho\,((x : v) \otimes (y : u)) = \rho\,(x : v)\,\rho\,(y : u)\,.$$

Its inverse is given by antisymmetrization: if $p\,(\rho) \neq 1$ then $\rho = \rho\,(1 + \operatorname{tr}\rho : w)$ where $A_w = 2\rho^{as}$, while if $p\,(\rho) = 1$ then $\rho = \rho\,(0 : v)$ for v any non-zero row of $I + \rho$.

Proof. *Multiplicativity*: with representatives $\alpha = (x, v)$ and $\beta = (y, u)$, the sandwich formula applied twice, then associativity and part 2 of the quaternion algebra theorem, give for any vector w

$$w\,\rho\,(\alpha)\,\rho\,(\beta) = \frac{\overline{\beta} \otimes (\overline{\alpha} \otimes w \otimes \alpha) \otimes \beta}{N\,(\alpha)\,N\,(\beta)} = \frac{\overline{\alpha \otimes \beta} \otimes w \otimes (\alpha \otimes \beta)}{N\,(\alpha \otimes \beta)} = w\,\rho\,(\alpha \otimes \beta)\,.$$

Surjectivity: a rotation with $p \neq 1$ is a cap (Section 7), hence $\rho\,(1 : w)$ with $A_w = A\,(\rho) = \frac{2}{1+\operatorname{tr}\rho}\rho^{as}$, which is the stated recipe after rescaling by $1 + \operatorname{tr}\rho$; a half turn is $\rho\,(0 : v)$, and its axis vector spans the row space of $I + \rho_{[v]} = \frac{2}{Q(v)}v^T v$.

Injectivity: by the half spread of a proportion, caps ($x \neq 0$, $p \neq 1$) and half turns ($x = 0$, $p = 1$) do not mix. On the chart $x \neq 0$, normalize $x = 1$: injectivity is part 5 of the theorem that caps are rotations. At infinity, distinct lines $[v]$ give distinct half turns by the half turns theorem of Section 4, and $(0 : v)\,, (0 : \lambda v)$ are the same proportion. ■

Corollary 67 *Writing $\mathbb{H}_{\mathbb{F}}^*$ for the group of non-null quaternions under $\otimes$, and $\mathbb{F}^*$ for the non-zero scalars inside it,*

$$SO\,(3, \mathbb{F}) \cong \mathbb{H}_{\mathbb{F}}^* / \mathbb{F}^*,$$

the non-null proportions under the quaternion product. Over $\mathbb{Q}$ and over ordered fields, where no null quaternions exist, this is all of P^3: $SO\,(3, \mathbb{Q})$ ***is*** *$P^3\,(\mathbb{Q})$* ***equipped with the projective quaternion product.***

The four breakdowns of Section 10 are now resolved: the half turns are the points $(0 : v)$ at infinity; the product is total on non-null proportions, with multiplicative quadrance; associativity holds unconditionally, being the associativity of the quaternion algebra; and the excluded anchors are the affine traces of null proportions—not missing rotations, but non-rotations.

11.3 The matrix model

Let ι denote a square root of -1: an element of $\mathbb{F}$ if -1 is a square there, and otherwise an adjoined one, with $\mathbb{F}[\iota] \equiv \mathbb{F}[X]/(X^2+1)$ a quadratic extension field. Write

$$J \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Theorem 68 (Matrix model) *The map*

$$\mu(x + t\mathbf{i} + r\mathbf{j} + s\mathbf{k}) \equiv xI + tJ + \iota rK + \iota sD = \begin{pmatrix} x+\iota s & -t+\iota r \\ t+\iota r & x-\iota s \end{pmatrix}$$

is an injective linear map from the quaternions over $\mathbb{F}$ into the 2×2 matrices over $\mathbb{F}[\iota]$, with

$$\mu(\alpha \otimes \beta) = \mu(\alpha)\,\mu(\beta), \qquad \det \mu(\alpha) = N(\alpha), \qquad \operatorname{tr} \mu(\alpha) = 2x,$$

and $\mu(\overline{\alpha})$ is the adjugate of $\mu(\alpha)$. In particular, for the rotation $\rho = \rho(x : v)$ with representative $\alpha = (x, v)$,

$$1 - p(\rho) = \frac{(\operatorname{tr} \mu(\alpha))^2}{4 \det \mu(\alpha)}.$$

If -1 is a square in $\mathbb{F}$ then μ is an isomorphism onto all of $M_2(\mathbb{F})$.

Proof. Linearity and injectivity are visible from the four independent entries. For multiplicativity it suffices, by bilinearity, to check the basis: $\mu(\mathbf{i}) = J$, $\mu(\mathbf{j}) = \iota K$, $\mu(\mathbf{k}) = \iota D$ each square to $-I$, and

$$\mu(\mathbf{j})\,\mu(\mathbf{i}) = \begin{pmatrix} \iota & 0 \\ 0 & -\iota \end{pmatrix} = \mu(\mathbf{k}), \quad \mu(\mathbf{k})\,\mu(\mathbf{j}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \mu(\mathbf{i}), \quad \mu(\mathbf{i})\,\mu(\mathbf{k}) = \begin{pmatrix} 0 & \iota \\ \iota & 0 \end{pmatrix} = \mu(\mathbf{j}),$$

matching the multiplication table, with the reversed products giving the negatives. The determinant is $(x+\iota s)(x-\iota s) - (-t+\iota r)(t+\iota r) = x^2+s^2+t^2+r^2$, the trace is $2x$, and the adjugate of $\mu(\alpha)$, obtained by swapping the diagonal entries and negating the others, is $\mu(x, -t, -r, -s)$. The half spread formula is $1 - p = x^2/N$ rewritten. When $\iota \in \mathbb{F}$, μ is an injective linear map between four dimensional spaces over $\mathbb{F}$, hence a bijection. ∎

For $\mathbb{F} = \mathbb{R}$, so that $\mathbb{F}[\iota]$ is the complex field, let

$$\tau_1 \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \tau_2 \equiv \begin{pmatrix} 0 & -\iota \\ \iota & 0 \end{pmatrix}, \qquad \tau_3 \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the Pauli matrices, usually written $\sigma_1, \sigma_2, \sigma_3$; we avoid that notation since σ_v denotes a reflection. Then $J = -\iota\tau_2$, $\iota K = \iota\tau_1$ and $\iota D = \iota\tau_3$, so that

$$\mu(\alpha) = xI + \iota(r\tau_1 - t\tau_2 + s\tau_3).$$

When $N(\alpha) = 1$ these are the usual $SU(2)$ matrices [11]. Thus the standard Pauli–spinor description of spatial rotations is already present in the matrix model; in the usual physics conventions the

three coordinates appear in a different order and with different signs. The projective formulation differs only in refusing to impose the normalization $N = 1$.

The identity

$$1 - p = \frac{\left(\operatorname{tr} \mu(\alpha)\right)^2}{4 \det \mu(\alpha)}$$

is especially useful because it survives unchanged in the relativistic companion [17]. There the basic algebra is $M_2(\mathbb{F})$ itself, and its projective action gives the rotations and boosts of the quadratic form of signature $(1, 2)$. The Euclidean and relativistic theories are therefore parallel projective constructions, distinguished by the underlying quadratic algebra.

Theorem 69 (Rotations over a finite field) *Over $\mathbb{F}_q$ with q odd, the quaternion algebra is isomorphic to $M_2(\mathbb{F}_q)$, the null proportions number $(q+1)^2$, and*

$$SO(3, \mathbb{F}_q) \cong PGL_2(\mathbb{F}_q) \qquad \text{of order } q\left(q^2 - 1\right).$$

Proof. If $q \equiv 1$ modulo 4 then -1 is a square and the matrix model gives the isomorphism directly. In general, the algebra has zero divisors: by the null vectors theorem of Section 2 there is a non-zero null vector v, and the pure quaternion $(0, v)$ is then null, hence not invertible by part 5 of the quaternion algebra theorem. A four dimensional quaternion algebra over a field is either a division algebra or isomorphic to the 2×2 matrices [19], [20], so the second alternative holds. Under the isomorphism N becomes the determinant, so null quaternions are singular matrices, of which there are $q^4 - (q^2 - 1)(q^2 - q) = q^3 + q^2 - q$ including zero; the non-zero ones give $(q^3 + q^2 - q - 1)/(q - 1) = (q+1)^2$ null proportions. The non-null proportions are the invertible matrices modulo scalars, that is $PGL_2(\mathbb{F}_q)$, of order $(q^2 - 1)(q^2 - q)/(q - 1) = q(q^2 - 1)$; and this is $SO(3, \mathbb{F}_q)$ by the projective rotation theorem. ■

Example 70 *Over $\mathbb{F}_5$ there are 36 null proportions and 120 rotations. Among them are the 25 half turns about the 25 non-null lines (the projective plane over $\mathbb{F}_5$ has 31 points, of which 6 lie on the null conic), and the parabolic rotations of Section 5.*

Remark 71 (Quaternions and dihedrons) *Suppose -1 is not a square in $\mathbb{F}$, and let $M \mapsto \overline{M}$ apply the automorphism $\iota \mapsto -\iota$ of $\mathbb{F}[\iota]$ to each entry. The image of μ consists of the matrices $\begin{pmatrix} a & -\overline{c} \\ c & \overline{a} \end{pmatrix}$, with $a = x + \iota s$ and $c = t + \iota r$; these are exactly the matrices M with $J\overline{M}J^{-1} = M$. Replacing this condition by $\overline{M} = M$ selects instead $M_2(\mathbb{F})$, which is the algebra of dihedrons used in the relativistic companion [17]; there a dihedron (x, t, r, s) has the matrix*

$$\mu_{rel}(x, t, r, s) \equiv \begin{pmatrix} x + s & r - t \\ r + t & x - s \end{pmatrix},$$

and $\mu(x, t, r, s) = \mu_{rel}(x, t, \iota r, \iota s)$. Thus the quaternion and dihedron algebras are two $\mathbb{F}$-forms of $M_2(\mathbb{F}[\iota])$. On vectors the change is $v \mapsto vS$, where $S = \operatorname{diag}(1, \iota, \iota)$ and $S^2 = \operatorname{diag}(1, -1, -1)$. This changes the relativistic quadrance $t^2 - r^2 - s^2$ into the Euclidean quadrance $t^2 + r^2 + s^2$. The companion paper develops the resulting Lorentz rotations and boosts in detail. Over finite fields the two algebra forms coincide; over $\mathbb{Q}$ one is a division algebra and the other is a matrix algebra.

11.4 Heritage of the quaternion product

The main formulas in this section are classical. Euler gave rational parametrizations of orthogonal matrices in the eighteenth century [2], and his four-square identity already contains the norm multiplication. Rodrigues gave the four component composition law in 1840 [5]. Related formulas occur in unpublished work of Gauss [4]. Hamilton then recognized the associative quaternion algebra behind them [8], and Cayley connected the algebra directly with rotations [7]. Modern treatments include [11], [21], [24].

Homogeneous quaternion coordinates, Rodrigues or Gibbs parameters [9], and the real identification $SO(3,\mathbb{R}) \simeq P^3(\mathbb{R})$ are therefore not new ingredients. The point here is their organization over the ground field itself. Over $\mathbb{Q}$ the projective quaternion group contains every rational rotation, while the norm-one rational quaternions do not. The next section measures this difference exactly.

12 The obstruction over the rational numbers

Every rotation is now a proportion $(x : v)$, and every proportion carries the square class of its quadrance $N(x, v) = x^2 + Q(v)$, which by the totality theorem of Section 11 is multiplicative. Write $(\mathbb{F}^*)^2$ for the group of non-zero squares, $\mathbb{F}^*/(\mathbb{F}^*)^2$ for the group of **square classes**, and $[c]$ for the square class of a non-zero c. So each rotation acquires an invariant in the group of square classes, and this invariant is computable in closed form.

Theorem 72 (Quadrance class of a rotation) *The map*

$$\chi : SO(3,\mathbb{F}) \longrightarrow \mathbb{F}^*/(\mathbb{F}^*)^2$$

sending a rotation to the square class of the quadrance of any of its proportion representatives is a well defined homomorphism, and

$$\chi(\rho) = [1 - p(\rho)] \ \text{ if } p(\rho) \neq 1, \qquad \chi\left(\rho_{[v]}\right) = [Q(v)].$$

Proof. Rescaling a representative by λ multiplies the quadrance by λ^2, so the square class is well defined, and multiplicativity is the totality theorem. For $p \neq 1$ take the representative $(1 : v)$ with anchor vector v: then $N = 1 + Q(v) = \frac{1}{1-p}$ by the half spread identities, and $\left[\frac{1}{1-p}\right] = [1-p]$. For a half turn take $(0 : v)$. ∎

Corollary 73 (Spinor norm) *For non-null vectors v_1 and v_2,*

$$\chi(\sigma_{v_1}\sigma_{v_2}) = [Q(v_1)\,Q(v_2)].$$

Proof. Write $d \equiv v_1 \cdot v_2$. If $d \neq 0$, the anchor of two reflections theorem gives the anchor vector $\frac{1}{d}v_1 \times v_2$, so by the Lagrange identity $\chi = \left[1 + \frac{Q(v_1 \times v_2)}{d^2}\right] = \left[\frac{Q(v_1)Q(v_2)}{d^2}\right] = [Q(v_1)\,Q(v_2)]$. If $d = 0$, then v_1 and v_2 are independent, $v_1 \times v_2$ has quadrance $Q(v_1)\,Q(v_2) \neq 0$, and $\sigma_{v_1}\sigma_{v_2}$ fixes $v_1 \times v_2$ and negates v_1 and v_2, which span $(v_1 \times v_2)^\perp$. By the half turns theorem it is $\rho_{[v_1 \times v_2]}$, with class $[Q(v_1 \times v_2)] = [Q(v_1)\,Q(v_2)]$. ∎

By the two reflections theorem this computes χ on every rotation, and identifies it with the classical **spinor norm** of the theory of quadratic forms [22], [19].

Theorem 74 (Surjectivity) *Over $\mathbb{Q}$ the homomorphism χ is onto the group $\mathbb{Q}^+/(\mathbb{Q}^*)^2$ of square classes of positive rationals $\mathbb{Q}^+$, which is infinite.*

Proof. For a positive rational $c = a/b$, the positive integer ab is a sum of four integer squares by the theorem of Lagrange, so $c = ab/b^2$ is a sum of four rational squares, giving a proportion of quadrance exactly c and a rotation of class $[c]$. Positivity is forced: over $\mathbb{Q}$ every quadrance $x^2 + Q(v)$ of a non-zero vector is positive. The classes $[2], [3], [5], [7], \ldots$ of the primes are pairwise distinct, so the group is infinite. ∎

Corollary 75 (The obstruction) *A rotation $\rho \in SO(3, \mathbb{Q})$ arises from a rational quaternion of quadrance 1 exactly when $\chi(\rho)$ is trivial, that is when $1 - p(\rho)$ is a square (respectively $Q(v)$ is a square, for a half turn). These rotations form a subgroup of infinite index in $SO(3, \mathbb{Q})$.*

Example 76 *The quarter turn $(t, r, s) \mapsto (t, -s, r)$ about the first coordinate axis is $\rho(1 : 1 : 0 : 0)$, with $p = \frac{1}{2}$ and $\chi = \left[\frac{1}{2}\right] = [2]$, non-trivial: no rational quaternion of quadrance 1 produces it, though the proportion $(1 : 1 : 0 : 0)$ produces it effortlessly. The 3–4–5 rotation $\rho(2 : 1 : 0 : 0)$ of Section 9, about the same axis, has $p = \frac{1}{5}$ and $\chi = \left[\frac{4}{5}\right] = [5]$, also non-trivial.*

Thus rational unit quaternions reach a subgroup of infinite index in $SO(3, \mathbb{Q})$. The projective quaternion representation has no such restriction. The difference is exactly the spinor norm.

12.1 Existence of rotations with prescribed axis and half spread

Theorem 77 (Existence) *Let v be non-null and $p \in \mathbb{F}$ with $p \neq 0, 1$. A rotation with axis $[v]$ and half spread p exists precisely when*

$$Q(v)\, p(1-p) \text{ is a non-zero square in } \mathbb{F},$$

equivalently when $Q(v)\, s$ is a square, where $s = 4p(1-p)$ is the spread. When the condition holds there are exactly two such rotations, mutually inverse, distinguished by their handedness. For $p = 1$ there is exactly one, the half turn $\rho_{[v]}$; for $p = 0$, over a field whose only null vector is zero, only the identity.

Proof. A rotation with $p \neq 0, 1$ has anchor A_w, and its fixed vectors form exactly the line $[w]$: the eigenvalue 1 of the characteristic polynomial $(X - 1)(X^2 - 2(1 - 2p)X + 1)$ is simple, since $X = 1$ in the quadratic factor gives $4p \neq 0$. So the axis condition forces $w = \lambda v$ with $\lambda \neq 0$, and the half spread condition, via the half twist, forces

$$\lambda^2 Q(v) = \frac{p}{1-p},$$

solvable for λ exactly when $\frac{p}{(1-p)Q(v)}$ is a square; multiplying by the square $((1-p)Q(v))^2$ identifies this class with that of $Q(v)\, p(1-p)$, and multiplying by 4 gives the spread form. The two solutions $\pm\lambda$ give the two anchor vectors $\pm\lambda v$, hence mutually inverse rotations, distinguished by handedness as in Section 8. The case $p = 1$ is the half turns theorem; for $p = 0$ the anchor A_w has $Q(w) = 0$, so $w = 0$ when the only null vector is zero. ∎

Example 78 (Rotations about a coordinate axis) *For $v = (1, 0, 0)$ the condition is that $p(1-p)$ be a square, and the solutions of $p(1-p) = \mu^2$ with $\mu \in \mathbb{Q}$ are exactly*

$$p = \frac{\lambda^2}{1+\lambda^2}, \qquad \lambda \in \mathbb{Q},$$

with $1 - 2p = \frac{1-\lambda^2}{1+\lambda^2}$: the achievable amounts of rotation about a coordinate axis correspond exactly to the Pythagorean triples, with the quarter turn at $\lambda = 1$ and the 3–4–5 rotation at $\lambda = \frac{1}{2}$.

Example 79 (No rational quarter turn about the diagonal) *For $v = (1, 1, 1)$ and $p = \frac{1}{2}$ the condition asks that $3 \cdot \frac{1}{4}$ be a square: it is not. There is no rational quarter turn about the main diagonal. But $p = \frac{3}{4}$ asks that $3 \cdot \frac{3}{16}$ be a square, and it is: the rotation is $c\left(A_{(1,1,1)}\right)$ itself, with half twist $Q = 3$. Which amounts of rotation an axis supports is governed by the square class $[Q(v)]$, an arithmetic fingerprint of the axis.*

Over $\mathbb{Q}$, for $0 < p < 1$ the quadratic factor $X^2 - 2(1-2p)X + 1$ has discriminant $-16p(1-p) < 0$, so a rational rotation other than the identity and the half turns has no eigenvalue in $\mathbb{Q}$ other than 1: in the language of the relativistic companion [17], every such rotation is elliptic.

Over a finite field $\mathbb{F}_q$ of odd characteristic there are two square classes. Under $SO(3, \mathbb{F}_q) \cong PGL_2(\mathbb{F}_q)$, the norm-one quaternions give $PSL_2(\mathbb{F}_q)$, an index two subgroup. For a fixed non-null axis v, the square condition on $p \in \mathbb{F}_q \setminus \{0, 1\}$ holds for

$$\frac{q - 2 - \eta(-Q(v))}{2}$$

values, where $\eta(c)$ is 1, -1 or 0 according as c is a non-zero square, a non-square or 0 in $\mathbb{F}_q$. (These values are the p-coordinates of the points of the conic $\left(p - \frac{1}{2}\right)^2 + Q(v)\, y^2 = \frac{1}{4}$ with $y \neq 0$; the conic has $q - \eta(-Q(v))$ points, two of them with $y = 0$, and the rest pair off as $\pm y$.) Thus the local obstruction is roughly half of the possible half spreads, with the exact count depending on the square class of the axis. Over $\mathbb{Q}$ the square-class obstruction has infinitely many values.

12.2 Ray proportions and spin

Suppose now that $\mathbb{F}$ is ordered. A **ray proportion** is a non-zero quaternion (x, v) taken up to multiplication by a positive scalar. Each projective proportion splits into two rays, represented by (x, v) and $(-x, -v)$.

The quaternion product respects positive rescaling. Since the Euclidean quaternion norm is positive on non-zero vectors over an ordered field, the ray proportions form a group. The map

$$\text{quaternion rays} \longrightarrow SO(3, \mathbb{F})$$

is surjective and two–to–one. Its kernel consists of the two scalar rays represented by 1 and -1.

Over $\mathbb{R}$ every positive ray has a unique representative of norm 1. The ray group is then the usual unit-quaternion group

$$SU(2) = Spin(3),$$

and the map above is the standard spin double cover of $SO(3, \mathbb{R})$.

Over $\mathbb{Q}$ the situation is more arithmetic. A positive rational quadrance need not have a rational square root, so a rational quaternion ray need not contain a rational unit quaternion. The rational points of the algebraic spin group, equivalently the norm-one rational quaternions, map only onto the kernel of the spinor norm χ. By contrast the ray group

$$\mathbb{H}_{\mathbb{Q}}^{*}/\mathbb{Q}^{+}$$

still gives a two–to–one cover of all of $SO(3,\mathbb{Q})$. Thus the familiar spin picture survives rationally, but its complete rational form is naturally projective rather than normalized.

The same distinction between projective and normalized spin data occurs in the relativistic companion [17], where the basic quadratic algebra is split and null directions are present from the start, and in the dihedron setting of [18].

13 Concluding remarks

13.1 What the projective formulation adds

The basic ingredients of this paper are classical: Rodrigues parameters, the Cayley transform, quaternions and homogeneous coordinates all have a long history. The new point is the way they fit together over the ground field.

The half spread

$$p = \frac{3 - \operatorname{tr}\rho}{4}$$

measures the amount of a rotation without introducing an angle. On the affine chart, the anchor is recovered directly from the antisymmetric part of the rotation matrix. Its quadrance is the half twist $p/(1-p)$, and the anchor product gives composition by vector arithmetic.

The affine chart has four exceptional features: half turns lie outside it, the anchor product has a forbidden denominator, associativity is only conditional there, and over some fields the quadric $Q = -1$ is excluded. Quaternion proportions remove all four at once. The result is the group isomorphism

$$SO(3,\mathbb{F}) \cong \mathbb{H}_{\mathbb{F}}^{*}/\mathbb{F}^{*},$$

where only non-null quaternion proportions are used. Over $\mathbb{Q}$ every non-zero quaternion is non-null, so this is all of $P^{3}(\mathbb{Q})$.

The projective viewpoint also exposes an arithmetic invariant which normalization hides. The square class of the quaternion quadrance is the spinor norm

$$\chi : SO(3,\mathbb{Q}) \longrightarrow \mathbb{Q}^{+}/(\mathbb{Q}^{*})^{2}.$$

It is onto. Rational norm-one quaternions therefore give only the kernel of χ, a subgroup of infinite index. Quaternion proportions give every rational rotation.

13.2 Physics and the relativistic parallel

Over the real numbers the matrix model is the usual Pauli-matrix model of quaternions. Norm-one representatives give $SU(2)$ and the standard spin double cover of $SO(3)$. The present formulation

keeps the same multiplication but treats the four quaternion coordinates homogeneously. This is useful whenever the data are naturally exact or algebraic.

The same construction has a direct relativistic parallel. In three dimensional Minkowski geometry the positive Euclidean quaternion algebra is replaced by the split 2×2 matrix algebra. Rotations and boosts are then described by projective matrix proportions, with null directions and parabolic transformations appearing naturally. The Euclidean and relativistic theories are two forms of the same 2×2 algebra after adjoining a square root of -1. The companion paper [17] develops this side of the story.

13.3 Computation

All formulas in the present theory use field operations. Composition of projective rotations is a four component quaternion product. The half spread is obtained from a trace. Away from half turns the anchor is obtained by antisymmetrization and one scalar division. Equality of proportions is checked by cross multiplication.

Over $\mathbb{Q}$ these operations may be carried out exactly. No square-root normalization is required. This does not by itself imply that every numerical implementation will be faster than a well-designed floating point quaternion library [23]; that depends on the application and the arithmetic being used. The projective representation is instead an exact alternative which retains the ground field throughout. It may be useful in symbolic computation, exact kinematics and situations where normalization is undesirable.

13.4 Open problems

Problem 80 *Classify the operations*

$$v \otimes u = \frac{v + u + \gamma\, v \times u}{1 + \delta\, v \cdot u}$$

with $\gamma \neq 0$ which are associative wherever defined. Conjugating the anchor product and its opposite by the rescaling $v \mapsto \lambda v$ gives the associative cases $\gamma = \pm\lambda$, $\delta = -\lambda^2$. Is $\delta = -\gamma^2$ necessary apart from degenerate operations?

Problem 81 *Count and classify the rotations of $\mathbb{F}_q^3$ by half spread, spinor norm and conjugacy type. In particular describe the parabolic classes on null axes and the distribution of half spreads on each square class of axis.*

Problem 82 *Study the kernel and image of the spinor norm on natural finitely generated subgroups of $SO\,(3, \mathbb{Q})$, and determine the possible spinor norms of rational rotations of finite order.*

Problem 83 *Replace $t^2 + r^2 + s^2$ by a general non-degenerate ternary quadratic form. The corresponding quaternion algebra should give the projective rotation group in the same way. The present Euclidean theory and the relativistic companion are the two basic examples.*

13.5 The projective viewpoint

A rotation can be described by a unit axis and an angle, or by four field elements up to a common non-zero scale. Over $\mathbb{R}$ these descriptions are readily converted into one another. Over $\mathbb{Q}$ they are not: normalization can leave the field, while projectivization does not.

This is the main lesson of the paper. The affine Rodrigues–Cayley theory is a chart. Quaternion proportions give the complete object. Over the rational numbers that object is exactly $P^3(\mathbb{Q})$ with its quaternion product, and its arithmetic is visible through half spread, quadrance class and spinor norm.

References

[1] L. Euler, *Formulae generales pro translatione quacunque corporum rigidorum*, Novi Commentarii Academiae Scientiarum Petropolitanae 20 (1776), 189–207.

[2] L. Euler, *Problema algebraicum ob affectiones prorsus singulares memorabile*, Novi Commentarii Academiae Scientiarum Petropolitanae 15 (1770) 1771, 75–106; Opera Omnia I-6, 287–315; Eneström E407.

[3] L. Euler, letter to C. Goldbach, 4 May 1748; in P. H. Fuss (ed.), *Correspondance mathématique et physique de quelques célèbres géomètres du XVIIIème siècle*, St. Petersburg, 1843.

[4] C. F. Gauss, *Mutationen des Raumes* (1819), Werke, Band 8, Göttingen, 1900, 357–362.

[5] O. Rodrigues, *Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace, et de la variation des coordonnées provenant de ces déplacements considérés indépendamment des causes qui peuvent les produire*, Journal de Mathématiques Pures et Appliquées 5 (1840), 380–440.

[6] A. Cayley, *Sur quelques propriétés des déterminants gauches*, Journal für die reine und angewandte Mathematik 32 (1846), 119–123.

[7] A. Cayley, *On the motion of rotation of a solid body*, Cambridge Mathematical Journal 3 (1843), 224–232.

[8] W. R. Hamilton, *Lectures on Quaternions*, Hodges and Smith, Dublin, 1853.

[9] J. W. Gibbs and E. B. Wilson, *Vector Analysis*, Yale University Press, New Haven, 1901.

[10] S. L. Altmann, *Hamilton, Rodrigues, and the quaternion scandal*, Mathematics Magazine 62 (1989), no. 5, 291–308.

[11] S. L. Altmann, *Rotations, Quaternions, and Double Groups*, Clarendon Press, Oxford, 1986.

[12] J. J. Gray, *Olinde Rodrigues' paper of 1840 on transformation groups*, Archive for History of Exact Sciences 21 (1980), 375–385.

[13] N. J. Wildberger, *Divine Proportions: Rational Trigonometry to Universal Geometry*, Wild Egg Books, Sydney, 2005.

[14] N. J. Wildberger, *Affine and projective universal geometry*, arXiv:math/0612499 (2006).

[15] G. A. Notowidigdo, *Rational trigonometry of a tetrahedron over a general metrical framework*, PhD thesis, UNSW Sydney, 2021.

[16] G. A. Notowidigdo and N. J. Wildberger, *Generalised vector products applied to affine and projective rational trigonometries in three dimensions*, arXiv:1903.08330 (2019).

[17] N. J. Wildberger, *Rational rotations of three dimensional Minkowski space: rotations, boosts and spinors*, preprint., preprint.

[18] N. J. Wildberger, *Dihedrons: an exceptional quadratic algebra and the rational geometry of relativistic rotations*, preprint.

[19] T. Y. Lam, *Introduction to Quadratic Forms over Fields*, Graduate Studies in Mathematics 67, American Mathematical Society, Providence, 2005.

[20] J. Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, Cham, 2021.

[21] J. H. Conway and D. A. Smith, *On Quaternions and Octonions: their Geometry, Arithmetic, and Symmetry*, A K Peters, Natick, 2003.

[22] J. Dieudonné, *La Géométrie des Groupes Classiques*, third edition, Springer, Berlin, 1971.

[23] M. D. Shuster, *A survey of attitude representations*, Journal of the Astronautical Sciences 41 (1993), no. 4, 439–517.

[24] J. B. Kuipers, *Quaternions and Rotation Sequences*, Princeton University Press, Princeton, 1999.